Gate B v2 and metrics: change log

What changed, why, what the re-run showed, and what the dashboard now shows.

Where this stands. All six phases are done on the branch gate-b-v2-and-metrics-consolidation. It is not pushed, as you asked, and not merged, so this shows on port 8101 only.

Cost and cash now come only from the card. The backtest always ran at your 0bp. The 30bp was added afterwards, to Gate v2 and a "net" column, by a rule in the brief that I applied over your Gate A ruling. That is removed, along with every other cost or cash number in the code.

Clenow's latest run (20261105-152026-31c4) now reads PASS on Gate v2, at your 0bp. Bootstrap P 0.054 clears the eased 0.10, but would miss the original 0.05. Factor alpha is amber at t 1.17. At 0bp nothing here tests what trading 10.3 times capital a year would cost: the sweep below does that.

Your rulings

You said	What I did
Use 0bp as set at Gate A. Remove all hard-coding of cost and cash	Card only: the gate, metrics, page and Excel use costs.spread_bps and universe.cash_asset. Details below
Approve, but make them easier	Bootstrap P eased from 0.05 to 0.10, and sub-period to 2 of 4 windows, none below -1.0. Pass at a true excess Sharpe of 0.5 rose from 42.6% to 57.1%. Alpha stays at t 2.0: any lower breaks the 5% false-pass target
Benchmark Sharpe	Where the deflated Sharpe bar starts. Dormant while the test is out
Remove the deflated Sharpe for now	Out of the gate, off in config. **So v2 no longer tests for selection luck**. The results table still reports it as a figure
INCOMPLETE is not the right way	Status is PASS or FAIL. A test with nothing to run on shows as not run and is listed beside the status
Good to proceed without those values	AUM, mandate limits, the book and product costs stay blank. Their tests show as not run
Archive, but keep the tearsheet for Excel	**Nothing archived:** the Excel calculation needs both tearsheet files, and the momentum rotation script produces that workbook
Hold the push	Not pushed

Cost and cash: what was removed

Where	Before	Now
Gate v2 and the "net" column	Charged a 30bp house cost over your 0bp	The run's returns as they are, at the card's cost. Gross shows beside net only if that cost is not zero
config/costs.yaml	house_round_trip_bp: 30	Only the appendix sweep grid. It never reaches a headline or a gate
config/fund.yaml	A 6% fallback I added yesterday	Removed. Only the appendix cash sweep. A card naming no rate gets Sharpe over zero, and the page says so
Clenow script	A dedicated 30bp comparator and two rows at 30bp and 100bp. Grids written in code	Removed. Grids read from config. A breakeven row replaces them
Rule text: CLAUDE.md, Gate A requirements, card drafting, universes, the model's prompts	"30bp is non-negotiable"	The card states a cost and a human confirms it at Gate A
Code defaults	Cost 5bp and 10bp in the engines and schema. Cash 6% in the pipeline, runner, cross-check and deciles, and 5% in the Excel tearsheet	No defaults. A missing value is an error, or comes from the card
Completeness and Gate A checks	A cost under 20bp counted as a fault	The check is that the card states one. Zero is a stated cost, and cost is always flagged for a human
Momentum rotation script	30bp, and 5% on the Excel sheet	It has no card, so you pass both when you run it
Simulations	6% and 30bp written in	In config/calibration.yaml, read by the simulations only. No strategy run reads them

A test now fails if a number like these is written back into live code.

Commits

Commit	What	Why
925289c	Freeze Gate B v1: version constants, manifest and workbook fields, baseline for every past run	So every later result can be compared with what v1 really said.
5139c54	One metrics module: every dashboard, workbook, report and gate reads ros/metrics/core.py	The same metric name gave different numbers depending on where you looked.
fab1b88	Gate v2: hard blocks, evidence scorecard, PASS/FAIL/INCOMPLETE, mandate file, card fields	Built so it could be simulated, before any threshold was chosen.
c471338	Calibrate Gate B by simulation: 8,000 strategies, real-data references, power curves and a proposal	So thresholds come from simulation, not from any one strategy.
b26130c	Fixes from the dashboard check: first-day trades count in turnover, older runs label their old Sharpe, runs stop copying core metrics	The dashboard check found these.
a73e1e9	Change log for the metrics consolidation and Gate v2 work, with the re-run and dashboard check	You asked for a record of the changes.
fb6bdc7	Approve easier Gate v2 thresholds, deflated Sharpe out, missing inputs show as not run	Your rulings, committed before any named strategy was scored on them.
0d5a58f	Wire Gate v2 into Lightyear: scored on every run, new Gate B tab and workbook sheet, past runs re-scored	So every run is scored on v2, on the page and in Excel.
6818b7f	Page, workbook and tests name where the cash rate came from	Superseded by the card-only rule below.
0b3ea3f	Reference document updated to v2: one formula per metric, the gate, its calibration, changes from v1; end-to-end gate tests	The document updated to v2.
684ef7b	Change log brought up to date: rulings, Gate v2 wired, v1 against v2 for every run, the Clenow comparison	You asked for a record of the changes.
ded8616	A run's cost and cash come only from its card: no 30bp, 6% or other number written into code	**Your ruling: the cost set at Gate A is the cost charged.**
1a76473	A value with no finite answer shows blank instead of breaking the Results page	My new breakeven row broke the page at 0bp.
0b260a2	Drop any row naming a house cost, re-score every run at its card's cost, documents updated	A leftover 30bp row from an earlier run.

84 files changed, 5476 insertions(+), 479 deletions(-), compared with the tag gate-v1. Tests: 645 passed, 1 skipped on the final code (570 passed before this work). ruff: no new errors against the original code.

Metric definitions: before and after

Metric	Before	Now
Sharpe	Four formulas: (CAGR - 6%) / vol on the tile, (CAGR - compounded cash) / vol in tables, (mean x 252 - 6%) / vol in Excel, mean excess / stdev in the deflated Sharpe	One: mean(r - c) / stdev(r - c) x sqrt(252)
Max drawdown	Positive in some places, negative in others	Daily values, positive
Drawdown duration	Calendar days	Trading days
Tracking error	Sample in one path, population in another	Sample stdev of daily gaps
Up / down capture	Daily means in one path, monthly in another	**Monthly means**. My choice: your brief named no frequency
95% CVaR	Rolling 21-day in one path, daily and negative in another	**Daily, positive loss**. My choice of frequency
Turnover	Mean x 252 in one path, sum / years in another	Sum / years
Bootstrap	Geometric Sharpe	The one Sharpe, same settings, plus a paired drawdown gap as evidence only

Added

What	Where
Core performance table: the same 16 metrics on every run, including Sortino and Calmar	Results tab; Core metrics sheet in Excel
Gate v2: hard blocks, scorecard, PASS or FAIL. v1 folded below for comparison	Gate B tab; Gate v2 sheet in Excel
Capacity. Code done, shows as not run until you supply AUM	Results tab, Gate B tab
Post-publication metrics	Results tab, Excel, scorecard
Diagnostics section for rows tied to one strategy's rules	Results tab, Excel Validation sheet
gate_version and metrics_version	manifest.json, workbook header, Gate B tab
Card fields: n_configs_selected_from, n_sensitivities, rules_fixed_in_print, fits_parameters	ros/cards/schema.py

Cut and kept

Item	Status
Mean drawdown, vol consistency, hit rate, monthly CVaR, the geometric and duplicate Sharpes	**Cut** from the pipeline report and metrics modules
Sterling, Treynor, Omega, Gain to Pain, tail ratio, adjusted beta, Jensen's alpha, extreme capture and the rest of your cut list	**Off Lightyear, still in the tearsheet**, because you need the Excel calculation. Its numbers can differ from the dashboard's
Geometric Sharpe	**Kept in one place**, so a paper's printed Sharpe compares on its own definition
validate/independent.py	**Kept separate on purpose:** the clean-room cross-check imports nothing from ros
universal_backtester/validation.py	**Not consolidated.** Its own bootstrap and deflated Sharpe remain, but no Lightyear path or gate uses them

Gate v2 and the calibration

Share passing	True Sharpe 0	0.5	0.75	1.0	Factor only	Best of 20	Best of 100
Status PASS (hard blocks)	6.9%	63.2%	87.4%	95.7%	73.4%	72.5%	97.3%
PASS and all green	4.5%	57.1%	84.7%	95.3%	2.6%	59.6%	95.9%

> **What the rulings cost.** PASS and all green meets the targets for no-skill and factor-only strategies. **Status PASS alone does not:** 73% of factor-only strategies and 6.9% of no-skill ones read PASS, because the scorecard does not block. With the deflated Sharpe out, 96% of the best of 100 random strategies get PASS and all green. And a card at 0bp is charged nothing, so PASS there says nothing about trading costs.

Thresholds came from simulations and were committed (fb6bdc7) before Clenow was scored on them. To be open about it: I had already seen Clenow's zero-cost P of 0.054 when I eased the bar past it. Formulas and curves: financial-metrics-and-gate-b-v2.pdf. gate-b-v2-calibration.pdf is the earlier proposal.

Clenow's latest run: v1 against v2

Test	Gate v1	Gate v2, at the card's 0bp
Status	**FAIL**, 1 of 5 blocking	**PASS**
Edge over the benchmark	Not tested	7.05%: PASS
Paired bootstrap P	0.054: fail at 0.05	0.054: PASS at 0.10
Worst window	-0.24 absolute Sharpe: fail at 0.30	Sharpe gap 0.11, 4 of 4 windows ahead: green
Factor alpha t	1.52 on 2 factors: fail at 2.0	1.17 on 7 factors: **amber**, does not block
Deflated Sharpe	0.00: fail	Out of the gate
After publication: edge	Not tested	5.51%: green
Data integrity	Not tested	PASS
Mandate, capacity, book	Mandate passed on the card's flag; no book	Not run: no values supplied

Appendix sweep: round-trip cost	0bp	10bp	20bp	30bp	50bp	75bp	100bp
CAGR	19.80%	19.25%	18.52%	17.95%	16.74%	15.16%	13.72%
Edge over the index	6.96%	6.42%	5.68%	5.12%	3.90%	2.33%	0.88%

On the page now	Strategy, at the card's cost	NIFTY 500
CAGR	19.80%	12.83%
Sharpe	0.83	0.49
Sortino	1.10	0.65
Calmar	0.88	0.34
Maximum drawdown	22.47%	38.30%
Drawdown duration (trading days)	547	490
Information ratio	0.49	–

Daily returns, trades and holdings are byte-identical to the earlier runs of this card. Turnover 10.33 times capital a year. Breakeven is beyond the sweep.

Every past run: v1 against v2

All 14 runs failed v1. Under v2, each at its own card's cost, 9 pass and 5 fail. Most of the passes are runs of the same Clenow card. Full detail: docs/gate_b/v1_vs_v2.csv.

Run	Strategy	v1	v2	Edge	P	Why it changed
20261006-181257-0394	Padhy 130/30	FAIL	**FAIL**	6.60%	0.468	Still fails: bootstrap P above 0.10
20261006-lee-swaminathan-1998	Lee-Swaminathan	FAIL	**FAIL**	-11.79%	0.941	Still fails: bootstrap P above 0.10
20261013-161034-6649	1_s2_0 (s0970389623000538)	FAIL	**PASS**	6.86%	0.047	Clears every hard block that could be run. Alpha (t 0.89) no longer blocks, and the deflated Sharpe is out
20261016-102923-bdc1	Clenow	FAIL	**FAIL**	9.05%	0.189	Still fails: bootstrap P above 0.10
20261021-104144-801e	Clenow	FAIL	**FAIL**	9.05%	0.189	Still fails: bootstrap P above 0.10
20261022-153434-39aa	Momentum 50 rebuild	FAIL	**FAIL**	4.25%	0.550	Still fails: bootstrap P above 0.10
20261030-124039-5511	Clenow	FAIL	**PASS**	8.19%	0.032	Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out

Run	Strategy	v1	v2	Edge	P	Why it changed
20261030-171327-0aa8	Clenow	FAIL	**PASS**	8.19%	0.032	Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out
20261030-173539-8fd8	Clenow	FAIL	**PASS**	8.19%	0.032	Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out
20261030-183103-3639	Clenow	FAIL	**PASS**	8.19%	0.032	Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out
20261030-191604-3399	Clenow	FAIL	**PASS**	8.19%	0.032	Clears every hard block that could be run. Alpha (t 1.53) no longer blocks, and the deflated Sharpe is out
20261102-105054-0427	Clenow	FAIL	**PASS**	7.05%	0.054	Clears every hard block that could be run. Alpha (t 1.17) no longer blocks, and the deflated Sharpe is out
20261105-132328-ab35	Clenow	FAIL	**PASS**	7.05%	0.054	Clears every hard block that could be run. Alpha (t 1.17) no longer blocks, and the deflated Sharpe is out
20261105-152026-31c4	Clenow	FAIL	**PASS**	7.05%	0.054	Clears every hard block that could be run. Alpha (t 1.17) no longer blocks, and the deflated Sharpe is out

Dashboard check

I opened the re-run in a real browser on the final code: **47 of 47** checks pass, with no script errors. Excel, after recalculation, matches the page on CAGR, volatility, Sharpe, drawdown, Sortino and Calmar. Run it again with scripts/gate_v2/check_dashboard.py.

Problems found and fixed

Problem	Cause	Fix
The gate charged 30bp on a 0bp card	A rule in the brief, which I applied over your Gate A ruling instead of raising the conflict	Card only
The Results page returned a server error at 0bp	**My bug.** Breakeven is infinite when the edge never reaches zero, and the page cannot carry that	Shows blank, and no value can do this again
Turnover read 10.33 and 10.26	**My bug.** The first day's purchases were dropped	Counted on the first day
Older runs said "Sharpe" but showed the old 6% number	My relabel	Labelled as the old formula, with a note to press Rebuild
7 rule-specific rows sat in Costs & execution	Written during the run	Lightyear moves them to Diagnostics
A factor equal to the strategy gave alpha t 17.6	A series regressed on itself	An identical factor is left out, and the row says so

Still open

Item	Why it matters
Claude's written text on this run still says "the 30bp house floor", in its strengths, weaknesses and notes	Written before the instruction changed. Future runs are told not to. Nothing enforces prose, as with "Evidence supports"
Mandate, capacity and book tests show not run	A PASS has not been tested on these. Filling in config/fund.yaml and config/mandate.yaml switches them on
The Gate B brief and "Evidence supports" are written from the v1 checklist	They can disagree with the v2 status above them. Labelled on the page
Each run can still add its own rows	"Rolling one-year behind" can show two values. Runs are told not to, but nothing enforces it
Older runs show v2 only after Rebuild	The table above scores them all, but their pages need Rebuild
config/calibration.yaml still holds 6% and 30bp	For the simulations only. No strategy run reads it
Your usual Lightyear on port 8100	It is on the old branch, and was not running when I last checked